

System User Guide & Administrator Reference

Deploying, operating, and securing the manufacturing workflow pipeline.

SYSTEM INFORMATION

Document Type:	Administrator & User Reference Manual
System Version:	v1.0.0 (Production Hardened)
Security Standard:	OWASP Top 10 Compliant
Release Date:	May 2026

1. Initial Credentials & Authentication

The Nobel Alloy system uses role-based access control (RBAC) to enforce security partitions between factory operators and system administrators. Below are the default credentials configured for deployment verification.

🔑 SUPER ADMINISTRATOR INITIAL CREDENTIALS

This account has full database authority, system-wide settings capabilities, operator creation rights, and audit trace viewership.

Email Address: `admin@nobel.com`

Initial Password: `admin123`

⚠️ SECURITY WARNING FOR PRODUCTION DEPLOYMENT

For maximum production security, change the administrator password immediately after initial login. The system supports custom seed environment parameters (`ADMIN_EMAIL` and `ADMIN_INITIAL_PASSWORD`) inside the backend `.env` file to generate unique initial admin credentials during database seeding.

Role-Based Access Control Structure

Role	System Authorization Level	Core Responsibility
Admin	Full access to CRUD on users, products, assignments, cards, bags, and system audit logs.	Factory floor oversight, analytics review, and setup.
Operator	Read-only assignments list. Write-only work log submission for target targets.	Fulfilling physical work and logging counters.

2. Core Operational Workflows

■ 1. Product Definition & Process Maps

Administrators define alloy products (e.g., specific alloy grades) and establish the sequential manufacturing process steps (e.g., Melting, Extrusion, QA, Packing) required to complete them.

■ 2. Operator Assignments

Target tasks are assigned to specific operators on specific shifts. Operators login to view their personalized assignment boards, complete operations, and log yields, counters, and reject details.

■ 3. Route Card Generation

Batches are grouped under Route Cards. Cards support selecting multiple calendar dates (interactive calendar pickers) to aggregate process operations from Master Data logs dynamically across multiple working days.

■ 4. Bag Bundling & Operator Traceability

Multiple Route Cards are packed into shipping Bags. The system dynamically runs aggregation functions to fetch contributing operators, process operations, and quantity yields, producing live operator traceability reports.

Analytical Reporting Engine

The system features real-time analytical reporting. Administrators can run calculations and export XLS/PDF reports for:

- Operator productivity and contribution reports
- Alloy production metrics and rejection rate summaries
- Process-specific cycle and throughput logs
- System audit history reports listing user operations

3. Production Security Protocols

The Nobel Alloy system has been hardened against security vulnerabilities to prepare for live production deployment.

Ø=Þáp Helmet Header Protection

Sets HTTP security headers including Content-Security-Policy (CSP), Frameguard to prevent clickjacking, and XSS filtering, while completely hiding backend signature headers like X-Powered-By.

Ø=Ý Secure Authentication (JWT & Bcrypt)

All passwords are hashed with 12 rounds of bcrypt. Tokens expire in 8 hours to minimize leakage windows, and user deletion endpoints prevent self-deletion.

#ó Rate Limiting

Login routes are locked to a maximum of 5 attempts per 15 minutes per IP to prevent brute-force attacks. Global endpoints are capped at 100 requests per 15 minutes.

Ø=Üæ Denial of Service (DoS) Prevention

Strict 10KB body size limits are enforced on incoming JSON payloads to block memory exhaustion attempts.

&™þ HARDENED SERVER INFRASTRUCTURE DETAILS

CORS Configuration:	Configured via CORS_ORIGIN environment variable
Proxy Trust Level:	Express trust proxy configured (Client IP forwarding)
Security Test Suite:	Automated test_security.js verification suite included

